

100 Days of Azure DevOps

Day 53 Handout — The agent builds, ACR receives, a human does not copy-paste a tag from Slack

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — laptop push vs pipeline publish (same image, different publisher)

Dimension	docker push from PC	Docker@2 buildAndPush
Who authenticates	Your user via az acr login	Service connection (AcrPush). Not the admin password in a variable
Tag	Whatever you typed. Friday :latest happens here	\$(Build.BuildId) plus optional latest pointer
Dockerfile	Whatever is on disk, dirty or not	What Git has on that commit. Multi-stage if the build needs a compiler
Scan	Optional local Trivy if you remember	Intro only today: a scan step can fail the job. Day 67 is the gate
Audit	No Build.BuildId, no commit SHA on the tag	Repo + build number is the receipt

Architecture B — decide tags and stages with a constraint

Constraint	Default	Change when
Publisher	Pipeline only for this proof	Never laptop-push a tag you will run in ACI/AKS
Tags	\$(Build.BuildId) required; latest optional pointer	Promoting by latest is not a release process
Dockerfile	Single stage until compile tools appear	Multi-stage: build in fat image, COPY --from=build into alpine/distroless
Connection	Docker Registry / ACR service connection, AcrPush on the RG or registry	Admin user enabled 'so the pipeline works' — that is a finding
Scan	Know Trivy/Defender-for-DevOps exist; do not block the series on a marketplace tour	Day 67: fail on secrets/critical CVEs with SARIF as the result format

Architecture C — pitfalls that publish the wrong bits on a Friday

Pitfall	What you see	Fix
Laptop is still the publisher	ACR shows a tag no build owns	Disable personal push for this repo; Docker@2 is the only writer
latest as identity	ACI pulls 'latest' that moved two commits ago	Substitute Build.BuildId into every run target
Secret in a pipeline variable	ACR admin password in Library	Service connection with AcrPush. Admin stays off
Build context includes .env	Image layers contain the secret	.dockerignore + never ARG for passwords
Scan as wallpaper	Green push, no CVE report	A scan intro is fine; a green push is not a security attestation

One-liner to remember

If humans push images by hand, humans will push the wrong tag on a Friday. Let the pipeline own the closet key and tag with the build, not a mood.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps org, project azure-100-labs. Microsoft-hosted ubuntu-latest. Personal ACR from Day 52.

- 1 Project Settings → Service connections → New → Docker Registry / Azure Container Registry. Use the personal ACR. Prefer identity over registry admin. Name it acr-connection.
- 2 Confirm the identity has AcrPush on that registry (or RG). If the wizard created Owner on the subscription, that is a finding — tighten before you call the lab done.
- 3 Add azure-pipelines.yml with pool vmImage ubuntu-latest and Docker@2 command buildAndPush, repository myapp, tags \$(Build.BuildId) and optionally latest. Dockerfile path from the repo root.
- 4 Run the pipeline. In ACR, confirm the Build.BuildId tag and digest. Do not docker push from the PC for this proof.
- 5 Optional intro: a Trivy (or similar) step that prints CVEs but does not yet fail the whole CD path. Scanning as a gate is Day 67.
- 6 Write docs/pipeline-push-day53.md: service connection name, build id tag, digest. If the job failed on auth, fix the identity — do not flip admin-enabled true.

Starter YAML — Docker@2 is the only publisher

```
trigger:
  - main
pool:
  vmImage: ubuntu-latest
variables:
  imageRepository: myapp
  tag: $(Build.BuildId)
steps:
  - task: Docker@2
    displayName: Build and push to ACR
    inputs:
      command: buildAndPush
      containerRegistry: acr-connection
      repository: $(imageRepository)
      Dockerfile: $(Build.SourcesDirectory)/Dockerfile
      tags: |
        $(tag)
        latest
```

Done checklist

- Pipeline, not the laptop, produced the tag in ACR
- Tag includes Build.BuildId; latest is at most a pointer
- Service connection has AcrPush — admin user still off
- docs/pipeline-push-day53.md has the digest

Tomorrow — Day 54

Azure Container Instances — a URL without a cluster gym membership